

A2 Level Mathematics (Edexcel IAL) — Formula Sheet

Edexcel International A2: WMA13/14 (P3, P4) + applied units. Builds on the AS sheet — those identities still apply.

PURE P3 — FURTHER TRIGONOMETRY & FUNCTIONS

Reciprocal functions	$\operatorname{cosec} \theta = 1/\sin \theta$; $\sec \theta = 1/\cos \theta$; $\cot \theta = 1/\tan \theta$
Identities	$1 + \tan^2 \theta = \sec^2 \theta$; $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
Addition formulae	$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$; $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
Double angle	$\sin 2A = 2 \sin A \cos A$; $\cos 2A = \cos^2 A - \sin^2 A$; $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
R addition	$a \sin \theta + b \cos \theta = R \sin(\theta + \alpha)$, $R = \sqrt{a^2 + b^2}$, $\tan \alpha = \frac{b}{a}$
Inverse trig domains	$\sin^{-1}: [-1, 1] \rightarrow [-\pi/2, \pi/2]$; $\cos^{-1}: [-1, 1] \rightarrow [0, \pi]$; $\tan^{-1}: \mathbb{R} \rightarrow (-\pi/2, \pi/2)$

P3 — EXPONENTIALS, LOGS & CALCULUS

Log laws	$\ln(xy) = \ln x + \ln y$; $\ln(x/y) = \ln x - \ln y$; $\ln(x^n) = n \ln x$
Exponential model	$y = ax^n \Rightarrow \ln y = \ln a + n \ln x$ (log-log, gradient n) $y = k \cdot b^x \Rightarrow \ln y = \ln k + x \ln b$ (ln-linear, gradient $\ln b$)
Chain rule	$dy/dx = dy/du \times du/dx$
Product & quotient	$(uv)' = u'v + uv'$; $(u/v)' = (u'v - uv')/v^2$
Standard derivatives	$e^{kx} \Rightarrow k \cdot e^{kx}$; $\ln kx \Rightarrow 1/x$; $\sin kx \Rightarrow k \cdot \cos kx$; $\cos kx \Rightarrow -k \cdot \sin kx$
Standard integrals	$\int e^{kx} dx = e^{kx}/k + c$; $\int 1/x dx = \ln x + c$; $\int \cos kx dx = \sin kx/k + c$

PURE P4

Partial fractions	$\frac{px+q}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$ — cover-up
Parametric equations	$dy/dx = (dy/dt)/(dx/dt)$; area = $\int y dx = \int y(t) \cdot x'(t) dt$
Implicit differentiation	differentiate both sides w.r.t. x ; $d/dx(y^2) = 2y \cdot dy/dx$
Binomial (any n)	$(1+x)^n$ expansion with $\binom{n}{r}$ coefficients; valid $ x < 1$
Integration by parts	$\int u dv = uv - \int v du$ (LIATE order: Log, Inverse, Alg, Trig, Exponential)
Volumes of revolution	about x -axis: $V = \pi \int y^2 dx$; about y -axis: $V = \pi \int x^2 dy$

VECTORS

Magnitude	$ a = \sqrt{x^2 + y^2 + z^2}$
Dot product	$a \cdot b = a b \cos \theta = x \cdot x + y \cdot y + z \cdot z$; $\perp$ iff $a \cdot b = 0$
Vector line	$r = a + t \cdot b$ (point + direction)

STATISTICS S2 / MECHANICS M2 (applied)

Normal approx to binomial	$X \sim B(n, p) \Rightarrow H \sim N(np, np(1-p))$ when $np > 5$ and $n(1-p) > 5$
Correlation & regression	$y = a + bx$; regression line passes through $(\bar{x}, \bar{y})$
Projectiles	$x: s = ut$ (no accel); $y: s = ut + \frac{1}{2}gt^2$; range $R = \frac{u^2 \sin 2\theta}{g}$
Work, energy, power	$W = Fd \cos \theta$; $KE = \frac{1}{2}mv^2$; $P = Fv$
Momentum/impulse 2D	resolve components; conserve momentum per axis